

ESM_3 Bootstrap Estimation for Network Scale-Up Using RDS Data

Introduction

When estimating the size or characteristics of a hidden population using the Network Scale-Up Method (NSUM), researchers typically assume a probability sample from the frame population. However, in many applied settings—including hard-to-reach populations—data are collected via Respondent-Driven Sampling (RDS). RDS introduces specific structural dependencies and inclusion probabilities that violate the assumptions of simple random sampling.

This presents a challenge: how can we correctly estimate uncertainty for NSUM estimates derived from an RDS sample? As shown in Feehan and Salganik (2016) and Salganik (2006), the NSUM estimator depends crucially on inclusion weights π_i , which must reflect the sampling design. When data are RDS-based, these weights are typically derived from known degree-based estimators such as RDS-I, or RDS-II or Gile's Successive Sampling (SS) weights.

To address this challenge, we propose a three-step bootstrap procedure for NSUM estimation using RDS data. This approach is flexible, modular, and applicable across several classes of NSUM estimators. It separates the issues of: 1. How to resample an RDS chain (Step 1), 2. How to recalculate sample-specific weights (Step 2), and 3. How to apply a chosen NSUM estimator (Step 3).

Step 1: Resampling the RDS Sample

We begin by resampling from the observed RDS sample in a way that mimics the original recruitment structure. Let the original sample be: $\mathcal{S} = \{i_1, i_2, \dots, i_n\}$ with recruitment chains and wave indicators. Let d_i denote self-reported degree for respondent i , and let the recruitment tree structure be encoded via seed/recruiter IDs.

Options for RDS Resampling

- Tree Bootstrap: Sample entire recruitment trees (originating from seeds) with replacement. This respects the hierarchical recruitment structure and allows design effect estimation (Salganik 2006).
- Successive Sampling Bootstrap (SSB): Sample with replacement according to inclusion probabilities derived from the SS model (Gile 2011).
- Neighborhood Bootstrap: Use ego-network topology to preserve recruitment ties and neighborhood structure (Yauck et al. 2022).

- Chain Bootstrap: Each bootstrap replicate samples chains or subchains with replacement, preserving recruiter–recruitee links. This is implemented in `surveybootstrap::rds.boot.draw.chain()` and used in studies like Weir et al. (2012).

Let $\mathcal{S}^{(b)}$ denote the sample drawn in bootstrap replicate (b) .

Step 2: Recalculating Weights

NSUM estimators require inclusion weights (π_i) or their inverses $(w_i = 1 / \pi_i)$. Because bootstrap samples differ in composition and recruitment pattern, these weights must be recomputed for each replicate.

General Structure

For each replicate (b) , construct: - $\mathcal{S}^{(b)}$: resampled respondent IDs - $d_i^{(b)}$: degree reports in replicate - $\pi_i^{(b)}$: estimated inclusion probabilities

Weighting Options

- RDS-II Weights (Volz and Heckathorn 2008): $w_i^{(b)} \propto \frac{1}{d_i^{(b)}}$
- SS Weights (Gile 2011): Incorporate sampling fraction and frame size (N_F) . Computed numerically via successive sampling approximation.

Let $\mathbf{X}_i$ denote covariates (e.g. traits, degree, indicator of hidden population membership), which are retained from the original data and passed to Step 3.

Step 3: NSUM Estimation on Resampled and Reweighted Data

Given a bootstrap sample $\mathcal{S}^{(b)}$, we estimate the size of the hidden population (N_H) using one of several NSUM estimators. All estimators rely on weighted sums of out-reports and degree.

(a) Generalized NSUM (GNSUM)

As described in Feehan and Salganik (2016), GNSUM estimates:

$$\hat{N}_H^{(b)} = \left(\frac{\sum_i \frac{y_{i,H}}{\pi_i^{(b)}}}{\sum_i \frac{d_i}{\pi_i^{(b)}}} \right) \cdot N_F$$

Where:

- $y_{i,H}$: number of known alters in the hidden population,
- d_i : degree (known network size, e.g. from Q13),
- (N_F) : size of the frame population (e.g. domestic workers in UK).

(b) GNSUM with Symmetric Visibility (for Hidden Members in RDS)

In our context, some RDS respondents are ex post identified as members of the hidden population. Under the symmetric visibility assumption, if $(i \in H)$, we assume the

probability that i knows j is the same as j knows i . This allows in-reports to be incorporated.

Let:

- $I_H(i) = 1$ if $i \in H$, 0 otherwise
- y_i^{in} : number of other hidden members who report knowing i

Then the estimator becomes:

$$\hat{N}_H^{(b)} = \left(\frac{\sum_i \frac{y_{i,H} + I_H(i)}{y_i^{\text{in}} \pi_i^{(b)}}}{\sum_i \frac{d_i}{\pi_i^{(b)}}} \right) \cdot N_F$$

(c) *Modified Basic Scale-Up (MBSU)*

This estimator adjusts for key reporting biases via three correction factors:

$$\hat{N}_H^{(b)} = \left(\frac{\sum_i \frac{y_{i,H}}{\pi_i^{(b)}}}{\sum_i \frac{d_i}{\pi_i^{(b)}}} \right) \cdot N_F \cdot \frac{1}{\delta \cdot \tau \cdot \rho}$$

Where:

- δ : Degree ratio ($\bar{d}_{H,F}/\bar{d}_{F,F}$) — ratio of average network degree of hidden population to frame population,
- τ : Visibility / true positive rate ($\bar{v}_{H,F}/\bar{d}_{H,F}$) — proportion of network edges correctly reported; measures transmission and reporting accuracy; captures visibility of hidden population to frame population,
- ρ : Popularity bias (relative visibility of hidden population members).

These values can be:

- Estimated using known subpopulations (e.g., alter groups with known size),
- Set by expert elicitation,
- Scanned in sensitivity analysis.

(d) *Model-Based NSUM*

Bayesian implementations (e.g. Maltiel et al. (2015)) model:

- Degree distributions,
- Reporting error,
- Group visibility,
- Hidden population size.

They yield a posterior distribution over (N_H) , and uncertainty is embedded within the model.

This step applies an NSUM estimator to the bootstrap sample $\mathcal{S}^{(b)}$ using the recalculated weights $w_i^{(b)}$ and responses $(y_{i,H})$, where $(y_{i,H})$ is the number of known contacts respondent i has in hidden population H .

Let (N_F) denote the frame population size (assumed known), and let (d_i) be the degree of respondent i .

Step 4: Aggregating Bootstrap Estimates

After (B) replicates:

$[\hat{N}_H^{(1)}, \dots, \hat{N}_H^{(B)}]$

We compute:

- Point estimate: $\bar{N}_H = \frac{1}{B} \sum_b \hat{N}_H^{(b)}$
- Standard error: sample SD
- 95% CI: empirical percentiles (e.g., 2.5%, 97.5%)